

IQ-9075 Evaluation Kit - Build And Flashing Guide

Overview

This explains how to build Yocto images for the Qualcomm IQ-9075 Evaluation Kit (QCS9075). Initialize the BitBake build environment and build a ready-made image target tailored to console, multimedia, proprietary multimedia, or edge AI use cases.

Prerequisites

- Host OS: Ubuntu 22.04 or 24.04 (x86_64)

```
1 sudo apt update && sudo apt install -y \  
2     gawk wget git diffstat unzip texinfo gcc build-essential \  
3     chrpath socat cpio python3 python3-pip python3-pexpect \  
4     xz-utils debianutils iputils-ping python3-git python3-jinja2 \  
5     libegl1-mesa libsdl1.2-dev pylint xterm python3-subunit \  
6     mesa-common-dev zstd liblz4-tool file
```

 The AWS build server and yocto-builder containers already have these installed.

Build Instruction

Source: [metropolis-io/iq9075-evk-yocto](https://github.com/metropolis-io/iq9075-evk-yocto)  Explain

```
1 # 1) Clone with all submodules  
2 git clone --recurse-submodules git@github.com:metropolis-io/iq9075-evk-  
yocto.git  
3 cd iq9075-evk-yocto  
4  
5 # 2) Initialize the build environment (must be sourced, not executed)  
6 source set-build-env  
7  
8 # 3) Build  
9 bitbake <image>
```

 You must run "source set-build-env" in every new shell session before invoking bitbake; otherwise, the environment and paths will not be configured.

Image	Description
metro-iq9075-console-image	Headless console with SSH, Qualcomm utilities, and security packages
metro-iq9075-multimedia-image	Full open source multimedia stack: Wayland, GPU, camera (libcamera), audio (PipeWire)

<code>metro-iq9075-multimedia-proprietary-image</code>	Multimedia stack with Qualcomm proprietary drivers: CAMX, Adreno, sensor binaries
<code>metro-iq9075-edge-ai-image</code>	Proprietary multimedia stack + OCI container runtime for edge AI workloads

Build Outputs

Built images land in `build/tmp/deploy/images/iq-9075-evk/`

Qualcomm layers source

✔ Using a populated DL_DIR and SSTATE_DIR can dramatically reduce build times by reusing previously fetched sources and compiled artifacts.

Qualcomm Linux 2.0RC2

Layer	Remote	Pinned Commit
bitbake	openembedded/bitbake Explain	b54600e
oe-core	openembedded/openembedded-core Explain	57fe3e4
meta-qcom	qualcomm-linux/meta-a-qcom Explain	cc27172
meta-qcom-distro	qualcomm-linux/meta-a-qcom-distro Explain	c1cfd67
meta-qcom-releases	qualcomm-linux/meta-a-qcom-releases Explain	986178f
meta-audioreach	AudioReach/meta-audioreach Explain	1e91ad6
meta-openembedded	openembedded/meta-a-openembedded Explain	0bc67b6

	Explain		
meta-security	https://git.yoctoproject.org/meta-security	9e6d962	
meta-selinux	https://git.yoctoproject.org/meta-selinux	27758bc	
meta-updater	 uptane/meta-update  r  Explain	fdc475d	
meta-virtualization	https://git.yoctoproject.org/meta-virtualization	9476782	

Qualcomm Linux 1.7

Flashing with QDL and fastboot

```

1 # Need root permission
2 sudo su
3 # make sure you are under: root@metropolis-ThinkPad-P16s-Gen-2:
4
5 # Confirm the device is in Qualcomm QDL mode
6 lsusb | grep -i qualcomm
7 # Expected: ID 05c6:9008 Qualcomm, Inc. Gobi Wireless Modem (QDL mode)
8
9 # Provision UFS
10 qdl --storage ufs prog_firehose_dds.elf
   partition_ufs/provision_1_2.xml
11 # Expected: UFS provisioning succeeded
12
13 # Flash SAIL NOR partitions
14 cd sail_nor/
15 qdl --storage spinor ../prog_firehose_lite.elf rawprogram0.xml
   patch0.xml
16 # Expected: SAIL partitions flashed successfully; 11 patches applied
17
18 # Return to the image directory
19 cd ..
20
21 # Flash UFS partitions
22 qdl --storage ufs prog_firehose_dds.elf rawprogram*.xml patch*.xml
23 # Expected: 104 patches applied; partition 1 is now bootable

```

Full QDL logging reference

```

1 root@metropolis-ThinkPad-P16s-Gen-
2 2:/home/metropolis/Workspace/IQ9075/metro-iq9075-flash-package-
   20260619063249/metro-iq9075-edge-ai-image# lsusb | grep -i qualcomm
3 Bus 002 Device 013: ID 05c6:9008 Qualcomm, Inc. Gobi Wireless Modem
   (QDL mode)
4
5 root@metropolis-ThinkPad-P16s-Gen-
6 2:/home/metropolis/Workspace/IQ9075/metro-iq9075-flash-package-

```

```
20260619063249/metro-iq9075-edge-ai-image# qdl --storage ufs
prog_firehose_ddr.elf partition_ufs/provision_1_2.xml
5 UFS provisioning succeeded
6
7 root@metropolis-ThinkPad-P16s-Gen-
2:/home/metropolis/Workspace/IQ9075/metro-iq9075-flash-package-
20260619063249/metro-iq9075-edge-ai-image# cd sail_nor/
8 root@metropolis-ThinkPad-P16s-Gen-
2:/home/metropolis/Workspace/IQ9075/metro-iq9075-flash-package-
20260619063249/metro-iq9075-edge-ai-image/sail_nor# qdl --storage
spinor ../prog_firehose_lite.elf rawprogram0.xml patch0.xml
9 waiting for programmer...
10 flashed "SAIL_HYP" successfully at 96kB/s
11 flashed "SAIL_SW1" successfully at 98kB/s
12 flashed "SAIL_HYP_BKUP" successfully at 96kB/s
13 flashed "SAIL_SW1_BKUP" successfully at 98kB/s
14 flashed "PrimaryGPT" successfully
15 flashed "BackupGPT" successfully
16 11 patches applied
17
18 root@metropolis-ThinkPad-P16s-Gen-
2:/home/metropolis/Workspace/IQ9075/metro-iq9075-flash-package-
20260619063249/metro-iq9075-edge-ai-image/sail_nor# cd ..
19 root@metropolis-ThinkPad-P16s-Gen-
2:/home/metropolis/Workspace/IQ9075/metro-iq9075-flash-package-
20260619063249/metro-iq9075-edge-ai-image# qdl --storage ufs
prog_firehose_ddr.elf rawprogram*.xml patch*.xml
20 waiting for programmer...
21 flashed "efi" successfully at 131072kB/s
22 flashed "system" successfully at 129529kB/s
23 flashed "overlay" successfully
24 flashed "PrimaryGPT" successfully
25 flashed "BackupGPT" successfully
26 flashed "xbl_a" successfully
27 flashed "xbl_config_a" successfully
28 flashed "shrm_a" successfully
29 flashed "PrimaryGPT" successfully
30 flashed "BackupGPT" successfully
31 flashed "xbl_b" successfully
32 flashed "xbl_config_b" successfully
33 flashed "shrm_b" successfully
34 flashed "PrimaryGPT" successfully
35 flashed "BackupGPT" successfully
36 flashed "PrimaryGPT" successfully
37 flashed "BackupGPT" successfully
38 flashed "aop_a" successfully
39 flashed "uefi_a" successfully
40 flashed "uefisecapp_a" successfully
41 flashed "uefisecapp_b" successfully
42 flashed "xbl_ramdump_a" successfully
43 flashed "dtb_a" successfully at 124245kB/s
44 flashed "dtb_b" successfully at 124245kB/s
45 flashed "tz_a" successfully
46 flashed "hyp_a" successfully
47 flashed "devcfg_a" successfully
48 flashed "cpucp_a" successfully
49 flashed "multiimgoem_a" successfully
50 flashed "multiimgqti_a" successfully
51 flashed "imagefv_a" successfully
52 flashed "toolsfv" successfully
53 flashed "PrimaryGPT" successfully
54 flashed "BackupGPT" successfully
55 flashed "aop_b" successfully
56 flashed "tz_b" successfully
57 flashed "hyp_b" successfully
58 flashed "devcfg_b" successfully
59 flashed "cpucp_b" successfully
```

```
60 flashed "multiimgoem_b" successfully
61 flashed "multiimgqti_b" successfully
62 flashed "imagefv_b" successfully
63 flashed "PrimaryGPT" successfully
64 flashed "BackupGPT" successfully
65 flashed "PrimaryGPT" successfully
66 flashed "BackupGPT" successfully
67 flashed "PrimaryGPT" successfully
68 flashed "BackupGPT" successfully
69 104 patches applied
70 partition 1 is now bootable
```